Microsoft Elevate

CAPSTONE PROJECT

Walmart Sales Performance Analysis using Power BI

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

Retail companies like Walmart generate large volumes of sales data daily across multiple branches, product lines, and customer segments.

However, without proper data analytics tools, it becomes difficult for management to monitor sales performance, identify profitable products, and understand customer purchasing behavior.

Walmart currently lacks a centralized analytical system that provides real-time insights into:

- Sales performance across cities and branches
- Product line profitability
- Customer purchase patterns
- Revenue trends over time

This lack of visibility makes strategic decision-making inefficient and limits the company's ability to optimize operations and improve sales performance.

PROPOSED SOLUTION:

To address this problem, a Power BI dashboard was developed to analyze Walmart sales data and provide actionable business insights.

The dashboard enables management to:

- Monitor overall sales performance
- Identify top-performing product categories
- Analyze customer purchasing behavior
- Evaluate branch performance
- Track revenue trends over time

Interactive filters and visualizations allow decision-makers to explore the data dynamically and make informed strategic decisions.

SYSTEM APPROACH:

The project was developed using Microsoft Power BI for business intelligence and data visualization.

Technologies Used:

• Power BI Desktop – Dashboard creation

• Power Query – Data cleaning and transformation

• DAX (Data Analysis Expressions) – Calculated measures

• CSV Dataset – Walmart sales transaction data

Data Preparation Steps:

• Data loading and validation

• Data type correction

• Feature engineering (Month, Day, Time of Day)

• Creation of measures such as Total Revenue and Profit

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection:**

Descriptive data analytics using **Power BI and DAX functions** (SUM, AVERAGE, COUNT) to analyze Walmart sales data and identify patterns in revenue, product performance, and customer behavior.

- **Data Input:**

Historical Walmart sales dataset including **branch, city, customer type, product line, quantity, total sales, date, payment method, and rating**. Additional fields such as **Time of Day, Day Name, and Month Name** were derived.

- **Training Process:**

Data was cleaned and transformed using **Power Query**. Calculated columns and measures were created using **DAX** to compute metrics like **Total Revenue, Quantity Sold, Profit, and Average Rating**.

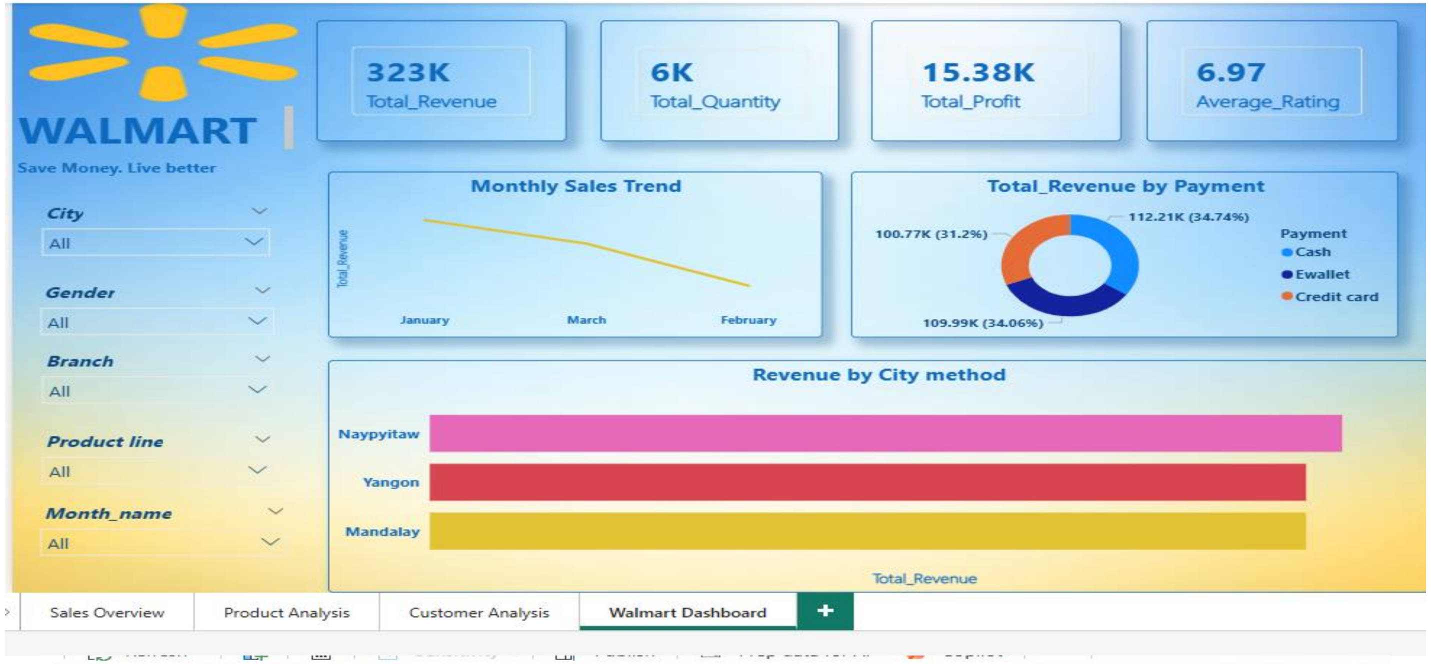
- **Prediction Process:**

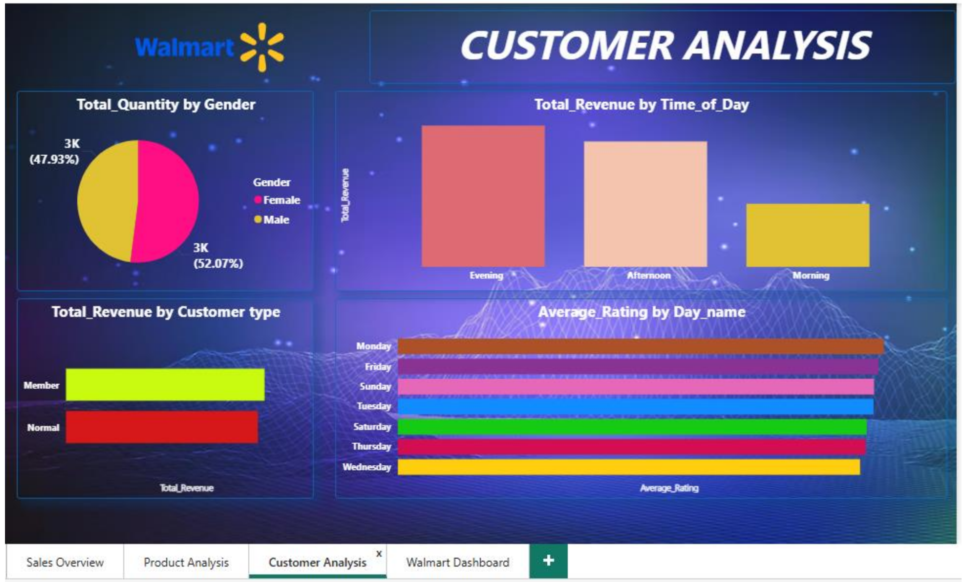
Interactive **Power BI dashboard** analyzes sales trends, product performance, and customer insights using visualizations and filters.

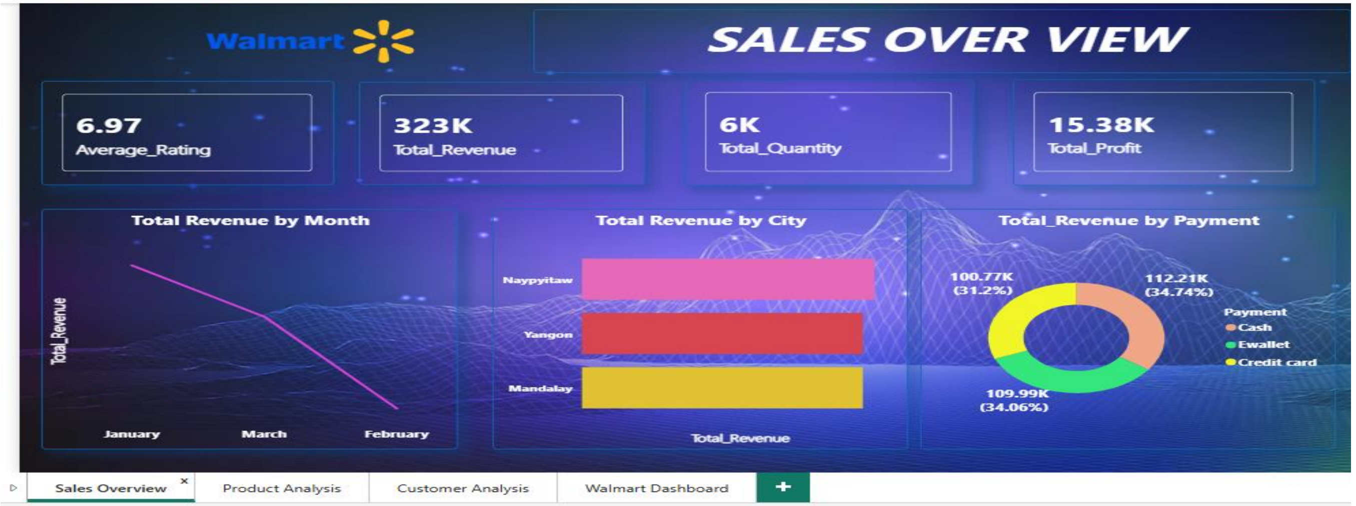
- **Deployment**

The dashboard is deployed in **Power BI** as an interactive reporting tool for **data-driven business decision-making**.

RESULT:


WALMART
Save Money. Live better
323K
Total_Revenue
6K
Total_Quantity
15.38K
Total_Profit
6.97
Average_Rating
City
All
Gender
All
Branch
All
Product line
All
Month_name
All
Monthly Sales Trend
Total_Revenue
January
March
February
Total_Revenue by Payment
100.77K (31.2%)
112.21K (34.74%)
109.99K (34.06%)
Payment
Cash
Ewallet
Credit card
Revenue by City method
Naypyitaw
Yangon
Mandalay
Total_Revenue
Sales Overview
Product Analysis
Customer Analysis
Walmart Dashboard


Walmart
CUSTOMER ANALYSIS
Total_Quantity by Gender
3K
(47.93%)
3K
(52.07%)
Gender
Female
Male
Total_Revenue by Time_of_Day
Total_Revenue
Evening
Afternoon
Morning
Total_Revenue by Customer type
Member
Normal
Total_Revenue
Average_Rating by Day_name
Monday
Friday
Sunday
Tuesday
Saturday
Thursday
Wednesday
Average_Rating
Sales Overview
Product Analysis
Customer Analysis
Walmart Dashboard


Walmart
SALES OVER VIEW
6.97
Average_Rating
323K
Total_Revenue
6K
Total_Quantity
15.38K
Total_Profit
Total Revenue by Month
Total_Revenue
January
March
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Total Revenue by City
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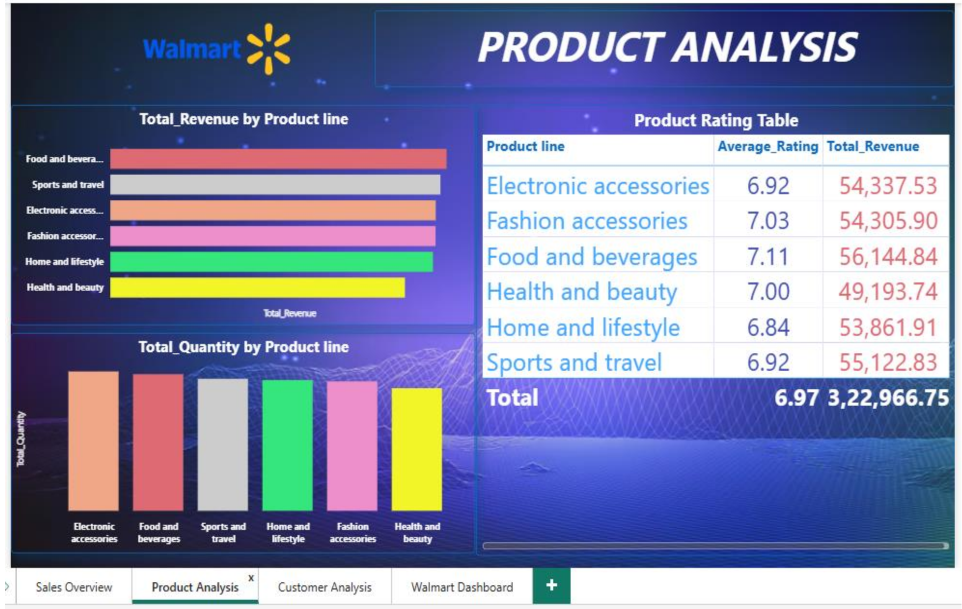
Walmart
PRODUCT ANALYSIS
Total_Revenue by Product line
Food and bevera...
Sports and travel
Electronic access...
Fashion accessor...
Home and lifestyle
Health and beauty
Total_Revenue
Product Rating Table
Product line | Average_Rating | Total_Revenue
Electronic accessories | 6.92 | 54,337.53
Fashion accessories | 7.03 | 54,305.90
Food and beverages | 7.11 | 56,144.84
Health and beauty | 7.00 | 49,193.74
Home and lifestyle | 6.84 | 53,861.91
Sports and travel | 6.92 | 55,122.83
Total | 6.97 | 3,22,966.75
Total_Quantity by Product line
Total_Quantity
Electronic accessories
Food and beverages
Sports and travel
Home and lifestyle
Fashion accessories
Health and beauty
Sales Overview
Product Analysis
Customer Analysis
Walmart Dashboard

CONCLUSION:

The Power BI dashboard provides a centralized platform for analyzing Walmart sales data and identifying important business insights.

The dashboard enables management to:

- Monitor sales performance effectively
- Identify high-demand product categories
- Understand customer purchasing behavior
- Improve operational decision-making

By leveraging data visualization and analytics, Walmart can make more informed and strategic business decisions.

FUTURE SCOPE:

Future enhancements for this system may include:

- Integration with real-time sales data
- Predictive analytics for demand forecasting
- Advanced customer segmentation analysis
- Expansion of the dashboard to include inventory and supply chain analytics
- Deployment through Power BI Service for enterprise-level accessibility

REFERENCES:

Walmart Sales Dataset – Kaggle

https://www.kaggle.com/c/walmart-recruiting-store-sales-forecasting

Microsoft Power BI Documentation

Data Analytics and Business Intelligence Best Practices

Git hub: Walmart_project_link

Thank You